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Evolutionary Generation of Targeted Images Using a Polygon-Based Genetic Algorithm (May 2025)

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ABSTRACT This paper presents a visual simulation system that evolves images using a classical Genetic Algorithm (GA) enhanced with modern perceptual quality metrics. Each image is encoded as a layered composition of semi-transparent polygons, ranging from triangles to hexagons, that gradually mutate to resemble a given target image. The system is tested on three distinct targets: a portrait of Albert Einstein, a stylized geometric tiger, and Vermeer’s Girl with a Pearl Earring. A composite evaluation strategy integrates both traditional similarity measures (mean squared error, peak signal-to-noise ratio, color histogram distance, and ΔE^* (CIE76) and perceptual metrics such as Structural Similarity Index (SSIM), Learned Perceptual Image Patch Similarity (LPIPS), and Fréchet Inception Distance (FID). Throughout the evolutionary process, the system generates intuitive visual outputs: animated GIFs showing generation-by-generation progress, final comparisons with the target image, metric plots across generations, and perceptual heatmaps. Our results show that the GA can produce visually recognizable approximations of complex images, with perceptual metrics guiding evolution toward solutions that align more closely with human visual perception. This project exemplifies a novel yet accessible fusion of classical evolutionary techniques with modern computer vision, emphasizing interpretability and offering a compelling educational case study in bio-inspired image optimization.

INDEX TERMS Computer vision, genetic algorithms, evolutionary computation, image processing, perceptual image quality, visual simulation.

I. INTRODUCTION

Genetic Algorithms (GAs) have been widely applied to evolve visual content through biologically inspired optimization. Early work by Sims [1] demonstrated how evolutionary methods, guided by human input, could generate complex computer graphics. More recent approaches use GAs to automatically approximate a given target image, reframing image reproduction as an optimization task.

These applications are not only visually compelling but also serve as effective educational tools, offering clear illustrations of evolutionary progress across generations. In typical GA-based image approximation, candidate solutions consist of layered simple primitives, such as translucent polygons, that combine to form increasingly

accurate representations. A notable example is Johansson’s “Evolved Mona Lisa” [2], which inspired many similar experiments using pixel-wise metrics like MSE or color difference.

While such metrics measure low-level similarity, they do not always reflect human visual perception. To address this, our work combines traditional metrics (MSE, PSNR, histogram distance, ΔE^* (CIE76) with perceptual measures such as Structural Similarity Index (SSIM) [4], Learned Perceptual Image Patch Similarity (LPIPS) [5], and Fréchet Inception Distance (FID) [6]. This multi-metric fitness function better aligns optimization with perceptual quality. The system is also designed for interpretability. It produces visual feedback across the evolutionary process, including

animations, metric plots, and perceptual heatmaps, making it ideal for educational and experimental use.

This paper presents an accessible yet effective framework that merges classical evolutionary computation with modern perceptual analysis. The remainder is structured as follows: Section II reviews related work, Section III outlines the system methodology, Section IV presents results and analysis, and Section V concludes with future directions.

II. RELATED WORK

The use of Genetic Algorithms (GAs) to evolve visual content has deep roots in evolutionary art. Sims's pioneering work [1] introduced interactive evolution in computer graphics, where users guided the process by selecting preferred images from procedurally generated patterns. This opened the door to many user-driven systems throughout the 1990s and early 2000s, allowing for the evolution of abstract visuals, textures, or even 3D forms. However, these systems relied on human input and were not designed to match a specific target image.

Later efforts explored automatic fitness functions to evolve images toward a reference. A well-known example is Johansson's "Evolution of Mona Lisa" [2], which used a mutation-based algorithm to combine 50 translucent polygons on a canvas. The system optimized pixel-level similarity, demonstrating that a recognizable portrait could emerge from noise. Bergen and Ross [3] followed a similar strategy with their JNetic platform, evolving vector-based graphics using circles, lines, and polygons. In both cases, fitness was computed using pixel-wise color distances, which do not always align with perceptual similarity as judged by humans. To address this gap, image processing research has introduced more sophisticated perceptual metrics. The Structural Similarity Index (SSIM) [4], proposed by Wang et al., compares structural patterns in luminance-normalized images and correlates better with perceived image quality than basic metrics like MSE or PSNR. More recently, Zhang et al. introduced the Learned Perceptual Image Patch Similarity (LPIPS) metric [5], which measures differences between deep neural network features, providing strong alignment with human similarity judgments. The Fréchet Inception Distance (FID) [6], another deep-feature-based metric, is widely used to evaluate generative models by comparing the feature distributions of real and generated images.

While these perceptual metrics are standard in modern image synthesis and computer vision, they are rarely integrated into evolutionary art systems due to their complexity and computational demands. Our work bridges this gap by incorporating SSIM, LPIPS, FID, and classical metrics (MSE, PSNR, ΔE^*) into a GA fitness function. This combined approach enables evolution not only toward low-level accuracy but also toward perceptual similarity. Moreover, we emphasize interpretability by visualizing how these metrics evolve across generations and influence the final output.

III. METHODOLOGY

This part is divided into the different phases taken in the algorithm development.

A. System Overview

We developed a modular image evolution system based on a classical Genetic Algorithm (GA). The pipeline follows a standard GA structure: population initialization, fitness evaluation, selection, crossover, and mutation over successive generations. The system is implemented in Python using libraries for image processing and machine learning, enabling the integration of perceptual metrics such as LPIPS and FID.

B. Image Representation

Each individual in the population encodes a candidate image composed of layered translucent polygons. Instead of representing each pixel, the image is built from a fixed number of polygons (e.g., 50), each defined by:

- A variable number of vertices (3 to 6),
- A set of 2D coordinates,
- An RGBA color (including transparency).

Polygons are rendered sequentially on a blank canvas, with their order either fixed or encoded. This low-dimensional, structured representation reduces the search space and naturally produces a low-poly aesthetic that is visually interpretable.

C. Genetic Operators

Each generation begins by evaluating the entire population based on the composite fitness function. Tournament selection is used to choose parents for variation.

Our system implements two crossover operators and one mutation strategy, all applied in the image space rather than directly on the polygon representation:

- **Blend Crossover.** A child is generated by computing a pixel-wise alpha blend of two parent images using a random mixing coefficient $\alpha \in [0, 1]$. This interpolation produces smooth combinations of visual traits from both parents.
- **Masked Crossover.** A random binary mask is generated across the image dimensions. For each pixel and channel, the value is taken from Parent A or B depending on the mask.
- **Mutation.** A child is created by duplicating a parent. With 50% probability, a new polygon is added to the image; otherwise, random noise is injected into the RGBA values to alter its appearance. The offspring is retained only if its fitness improves.

These simple operators promote gradual exploration and variation while maintaining computational efficiency. No polygon-level structural crossover is performed.

D. Fitness Function

During evolution, the fitness of each individual is computed solely using the average perceptual color difference between the candidate and target image. Specifically, we use the CIE76 ΔE^* metric, computed pixel-wise in the LAB color space:

$$F(\text{candidate}) = \text{mean}(\Delta E^*(\text{target}, \text{candidate}))$$

This metric captures perceptual differences in color more reliably than raw RGB distance, while remaining computationally efficient. Individuals with lower ΔE^* values are considered more fit.

While additional metrics such as MSE, SSIM, LPIPS, histogram distance (Wasserstein Distance), and FID are computed after training for analysis, they are not part of the fitness function during evolution.

E. Modularity and Implementation

The system is modular by design. Core components (polygon representation, variation operators, and evaluation) are encapsulated and can be modified independently. For instance, new shape primitives or evaluation metrics can be added without changing the overall GA structure.

Standard image quality metrics such as SSIM and PSNR are computed using open-source libraries, while LPIPS and FID are obtained via pre-trained PyTorch models. Although these metrics can be computationally intensive, our system is structured to allow flexible control over their frequency and scope. In practice, all metrics are evaluated for every individual to ensure consistent post-analysis. The system also saves snapshots, logs metric values, and stores intermediate visualizations to support interpretability.

F. Experimental Setup

We tested the system on three 256×256 target images: (1) a grayscale portrait of Albert Einstein, (2) a flat-color geometric tiger illustration, and (3) Vermeer's *Girl with a Pearl Earring*. These images vary in detail, texture, and color complexity, providing a diverse benchmark.

Each run used a population of 50 individuals and between 5,000 and 20,000 generations, depending on the difficulty of convergence. GA parameters (mutation rates, operator probabilities, etc.) were kept constant across experiments.

IV. RESULTS AND DISCUSSIONS

A. Evolutionary Process Visualization

The evolutionary runs produced a series of intermediate images that show the gradual improvement of the population's best candidate toward the target. We captured these in evolutionary progress animations (GIFs), a sequence of which is illustrated in **Figure**.

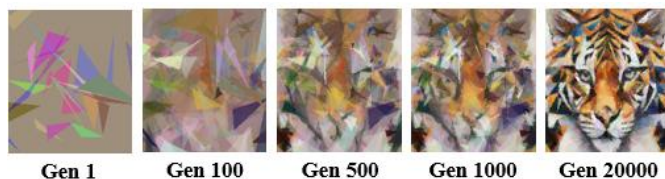


Figure 1. Evolutionary progress of the best individual across generations for the geometric tiger image.

Initially (e.g., generation 1), the image was an unstructured arrangement of colorful polygons with no discernible pattern

or meaning. By generation 100, some dominant color regions had begun to emerge, particularly in the central area of the canvas, where light and dark tones started to cluster.

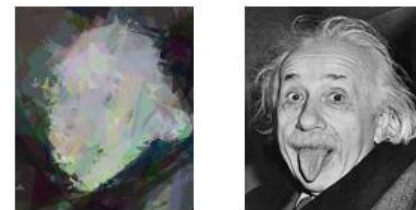
At generation 500, the early silhouette of the tiger's face became noticeable. Orange and black polygonal patches began to align vertically, suggesting the location of the eyes and the nose bridge. Although the shapes were still rough and edges blurry, the general layout of features was in place.

By generation 1000, the structure had refined further: the eyes had converged toward symmetric dark regions, the muzzle area had lightened, and the background began to separate from the tiger's outline. From that point on, subsequent generations focused on improving edge sharpness and contrast, especially around the tiger's facial stripes.

By generation 20,000, the result closely resembled the target image. The polygon shapes were layered in a way that produced strong visual coherence.

B. Final Evolved Images vs Targets

The final evolved images for the three targets are shown in **Figure**.



Final Image **Target Image**

Figure 2a. Einstein Example



Final Image **Target Image**

Figure 2b. Tiger Example



Final Image **Target Image**

Figure 2c. Girl Example

Figure 2. Final evolved images vs original targets

The Einstein portrait (Figure a) is clearly identifiable: the outline of Einstein's face and hair, and even features like the eyes and the characteristic mustache, are present. On close inspection, the evolved image has an abstract quality but from a normal viewing distance, the likeness is evident.

The *geometric tiger* result (Figure b) is accurate in terms of shape placement; the polygons effectively act as pieces of a jigsaw, recreating the tiger's form. Minor differences can be observed at the boundaries of stripes where the polygon edges cannot perfectly reproduce the smooth curves of the original vector graphic, but the color blocks and overall composition match almost exactly.

For the *Girl with a Pearl Earring* (Figure c), the evolved image captures the face, the dark background, and the iconic blue headscarf with yellow tassel. The facial features in the evolved version are slightly more cartoonish or smoothed (likely due to polygon limitations), yet the lighting and color tones are impressively close to Vermeer's original.

It is useful to know that each final image inherits a certain stylization from the polygon representation. This is not an imperfection per se, but rather an outcome of the representation choice: the Einstein image looks like a mosaic or stained-glass version of the photograph, and the *Girl with a Pearl Earring* result appears as if it were a modern digital art rendition of the painting. In some contexts, this could be considered a desirable creative effect. In our evaluation, we focus on how well these images approximate the targets in various metrics.

C. Quantitative Metrics Analysis

To quantitatively assess the quality of the final evolved images, we computed several standard image similarity metrics for the two completed experiments: Einstein and the geometric tiger. These include pixel-wise (MSE, PSNR), structural (SSIM), perceptual (ΔE^* , LPIPS, FID), and distributional (histogram) metrics. The results are summarized in Table I.

TABLE I
FINAL METRIC VALUES FOR THE BEST EVOLVED INDIVIDUALS

Metric	Einstein	Tiger
<i>MSE</i>	862.33	1132.29
<i>PSNR (dB)</i>	18.77	17.59
<i>SSIM</i>	0.3433	0.6752
ΔE^* (CIE76)	13.40	17.80
<i>Histogram Distance</i>	7.97	13.28
<i>LPIPS</i>	0.5954	0.2882
<i>FID</i>	442.44	271.06

These results show clear differences between the two target types. The Einstein image yielded lower MSE and ΔE^* , indicating better pixel-level and color fidelity. However, its SSIM and LPIPS scores were worse than those of the tiger, reflecting a lower structural and perceptual match. This can be attributed to the complexity of the Einstein portrait which is harder to approximate.

The tiger image, in contrast, had higher MSE and ΔE^* values, but significantly outperformed in SSIM (0.6752) and LPIPS (0.2882). This reflects that although the pixel-wise differences were larger, the overall layout and perceptual similarity were better captured. The flat-color regions and geometric layout of

the tiger made it more compatible with polygonal reconstruction.

PSNR values were in the 17–19 dB range for both runs, a typical level for low-resolution, stylized image approximations. FID scores also supported these observations: the tiger result had a lower FID (271.06) than Einstein (442.44), indicating a closer match in deep feature space.

Importantly, no single metric alone captures the full quality of the evolved images. MSE and PSNR favor smoothness, while SSIM and LPIPS reward structure and texture. FID emphasizes realism and coherence in a feature-based sense.

These results also confirm that the ΔE^* -based fitness function is effective in guiding convergence, but that full evaluation of output quality requires post-hoc analysis using perceptual and distributional metrics.

D. Perceptual Difference Heatmaps

To better understand where differences remained between the evolved images and their targets, we computed perceptual error maps based on ΔE^* (CIE76) in the Lab color space. Figure 3 presents visualizations for the Einstein portrait and the geometric tiger image. In each case, we show the original target, the best evolved image, and a corresponding heatmap that highlights per-pixel perceptual differences, with warmer tones indicating larger color discrepancies.



Figure 3a. Einstein Example



Figure 3b. Tiger Example

Figure 3. Perceptual difference heatmap

In the Einstein case (Figure 3a), the heatmap reveals that most of the facial region was accurately approximated. The highest ΔE^* values are concentrated around small-scale details such as the eyeglass rims, the mouth contours, and the wrinkles in the face. These areas are particularly sensitive to small deviations in edge placement and color blending, which are difficult to replicate with limited polygon count and transparency blending. The hair region and background also exhibit scattered error due to texture complexity.

In the tiger image (Figure 3b), the heatmap shows sharper error localization along the stripe boundaries, consistent with the geometric nature of the original. The GA correctly placed the major color blocks, but the polygon approximation struggled to match the curvature and precision of the original pattern edges.

V. CONCLUSION AND FUTURE WORK

We presented a visual simulation system that uses a classical Genetic Algorithm (GA) to evolve images toward a target, using a representation based on layered translucent polygons. The system was tested on diverse image types, a grayscale photograph, a stylized digital graphic, and a classical painting, demonstrating its flexibility and effectiveness.

Although the GA employed a traditional optimization structure, it was enhanced through the use of modern perceptual evaluation metrics, including ΔE^* (CIE76), MSE, PSNR, SSIM, LPIPS, histogram distance, and FID. While only ΔE^* was used as the optimization fitness in our implementation, the remaining metrics were computed post hoc to assess and interpret the quality of the evolved images from multiple perspectives.

This hybrid approach, combining a classical bio-inspired algorithm with deep learning-based perceptual metrics and visual diagnostics such as difference heatmaps, highlights an underexplored yet pedagogically valuable strategy. It emphasizes visual interpretability and exposes students to both traditional evolutionary computation and contemporary image quality assessment techniques. In contrast to typical coursework that relies on basic error metrics, this project illustrates how integrating perceptual tools can yield results that align more closely with human judgment.

Our results show that even with a limited representation the GA was capable of generating compelling approximations of complex images. The system captured global structure and color distribution effectively, with residual errors localized to fine detail regions such as curved edges and textured transitions.

Looking ahead, several directions may improve the system's performance and generality. First, allowing a variable number of polygons or introducing richer primitives could enhance detail and reduce approximation error. Second, employing a true multi-objective evolutionary algorithm would allow simultaneous optimization across metrics.

Finally, we see this system as a promising educational tool. By allowing users to evolve images interactively while observing the effects of metric selection and parameter tuning, it offers a hands-on learning experience in both bio-inspired optimization and computer vision.

In summary, our work demonstrates that classical GAs, when coupled with modern perceptual analysis and visualization, remain a powerful and interpretable tool for image synthesis. This fusion of old and new serves as a stepping stone for further interdisciplinary research in evolutionary image generation.

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